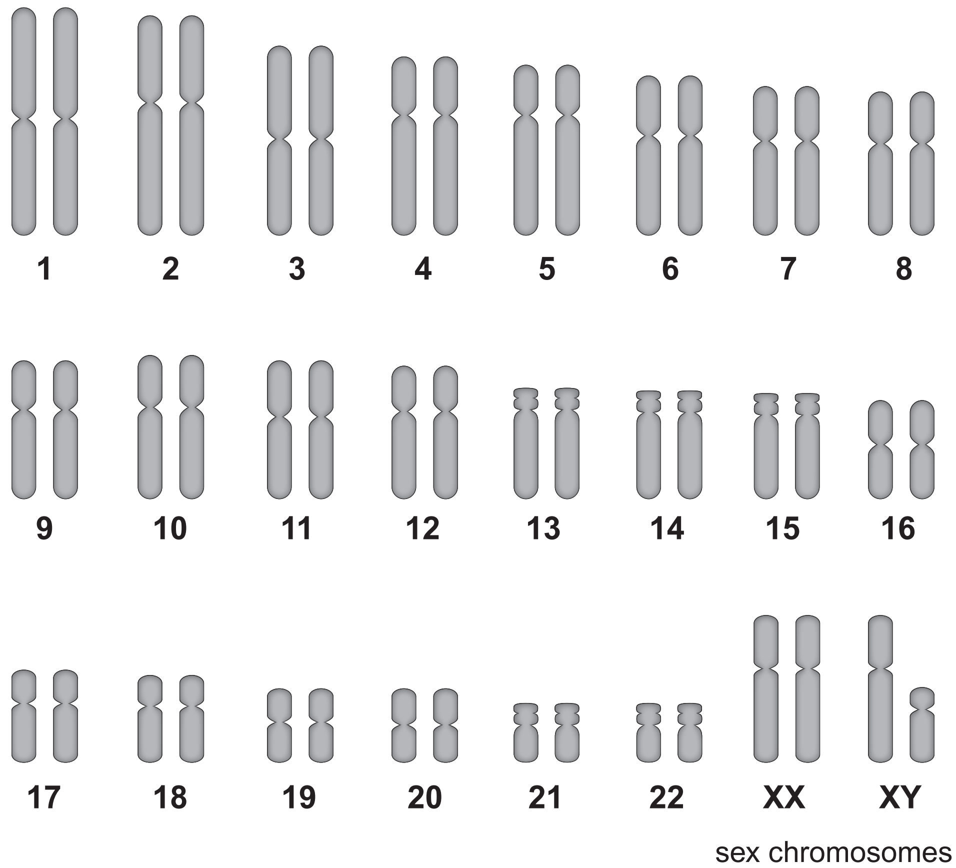
Examiner only

*Answer* **all** *questions.*

**1.** *(a)* The image shows human chromosomes.




(i) State the number of pairs of chromosomes present in a single body cell of a human. [1]

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(ii) State which sex chromosomes are present: [1]

in body cells of a man; ............................................................

in body cells of a woman. ............................................................

3400UB01
03

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(iii) Complete the Punnett square below to show the sex chromosomes in the gametes of a male and female parent and in their possible offspring. [2]

Male Parent

Female Parent

| Gametes | | |
|---|---|---|
| | | |
| | | |

*(b)* Height in humans is controlled by many genes as well as by environmental factors. Females usually reach their full height by age 18 and males by 25.

Students investigated the heights of male and female airline cabin crew. All cabin crew must be between 1.58 m and 1.90 m in height and between 18 and 45 years of age.

The students said that they expected the males to be taller than the females.

They collected their data by selecting seven males and seven females at random and asking them to state their heights. They gave their heights in feet and inches and the students converted the data into metres.

The results, expressed to two decimal places, are shown in the table below.

| Heights of females (m) | Heights of males (m) |
|---|---|
| 1.80 | 1.84 |
| 1.78 | 1.80 |
| 1.83 | 1.72 |
| 1.68 | 1.70 |
| 1.75 | 1.61 |
| 1.82 | 1.81 |
| 1.69 | 1.73 |
| mean height = ................ | mean height = 1.74 |

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(i) **Complete the table** by calculating the mean height for females. [2]
*Space for working.*

(ii) I. State the hypothesis that the students were testing in their investigation. [1]

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II. State whether the results of the investigation support their hypothesis, giving the reason for your answer. [1]

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III. Give **one** way in which the strength of the evidence could be improved. [1]

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*(c)* State **one** source of inaccuracy in the method. [1]

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10